

Ellipses



Standard form

- 1 For the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$, find:
 - a the vertices
 - b the foci
 - c the endpoints of the minor axis
 - d the eccentricity
 - 2 Write the standard-form equation of the ellipse with foci $(0, \pm 3)$ and vertices $(0, \pm 5)$.
 - 3 Consider the ellipse $9x^2 + 4y^2 = 36$.
 - a Write the equation in standard form.
 - b State the lengths of the major and minor axes.
 - c Find the coordinates of the foci.
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Applications

- 4 A whispering gallery has an elliptical ceiling 40 m long and 24 m high at the center (so $a = 20$ and $b = 12$, with the major axis horizontal). Two people stand at the foci. How far apart are they, to one decimal place?
 - 5 An ellipse centered at the origin has a horizontal major axis of length 12 and passes through the point $(3, 2\sqrt{3})$. Find its equation.
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Ellipses centered away from the origin

- 6 For the ellipse $\frac{(x-2)^2}{16} + \frac{(y+1)^2}{9} = 1$, find:
 - a the center
 - b the vertices
 - c the foci
 - 7 Complete the square to write $4x^2 + y^2 - 8x + 6y + 9 = 0$ in standard form, and state the center.
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Eccentricity and orbital applications

- 8 Compute the eccentricity of each ellipse, and state which is more circular (closer to $e = 0$):
 - a $\frac{x^2}{100} + \frac{y^2}{36} = 1$
 - b $\frac{x^2}{25} + \frac{y^2}{24} = 1$

- 9 A planet's elliptical orbit has semi-major axis $a = 150$ million km and eccentricity $e = 0.02$. Find the distance from the center to a focus, c , and hence the closest approach distance to the sun ($a - c$), to two decimal places (in millions of km).